PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 12 · Introduction to Machine Learning Models

TUESDAY NOV 10 AND THURSDAY NOV 12, 2026

01

MACHINE LEARNING MODELS

MACHINE LEARNING MODELS

1. **Regression**

   Predicts a number from a weighted sum of the inputs. Weeks 5 and 6.

2. **Decision tree.**

   Asks yes or no questions about the inputs to categorize the results

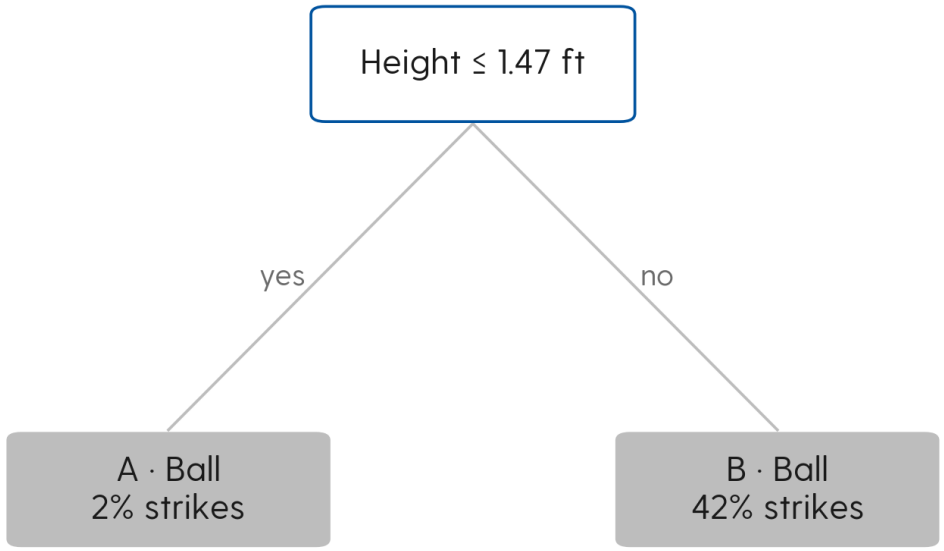
3. **Random forest.**

   Mulitple decision trees, each with one vote



A BATTER TAKES A PITCH. WHAT WOULD YOU WANT TO KNOW ABOUT THE PITCH TO PREDICT THE CALL?

ONE LEVEL TREE STRIKE ZONE

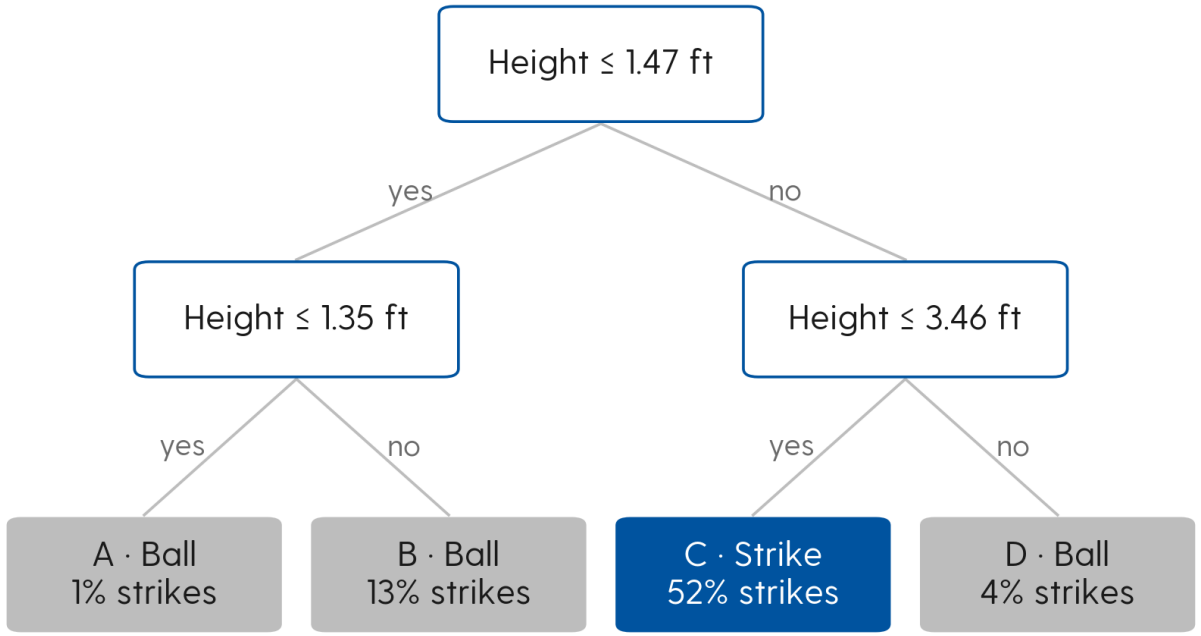


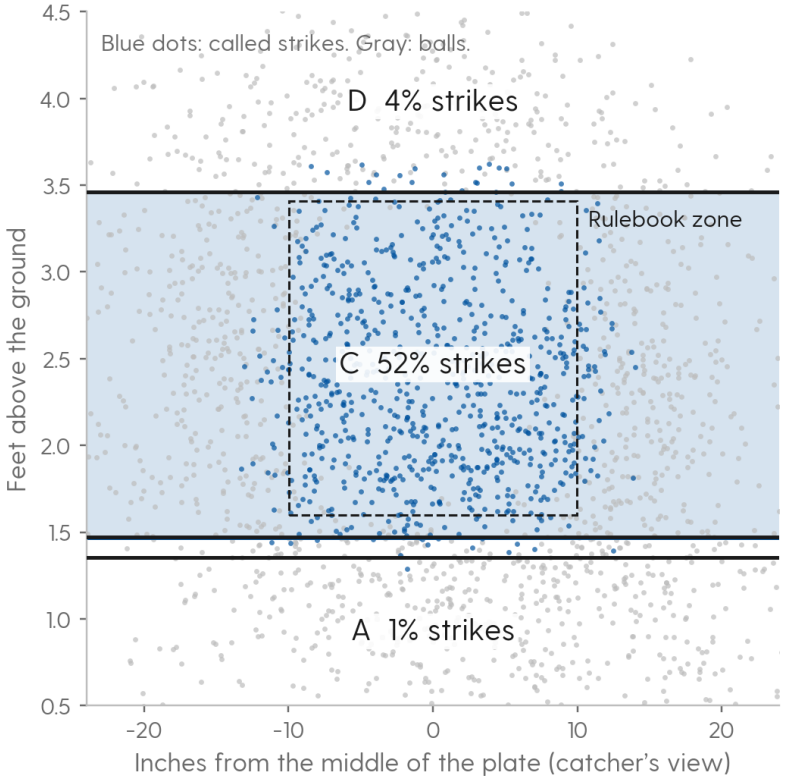


Every pitched is categorized as a ball. Accurate 68.3% of the time.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

TWO LEVEL TREE STRIKE ZONE

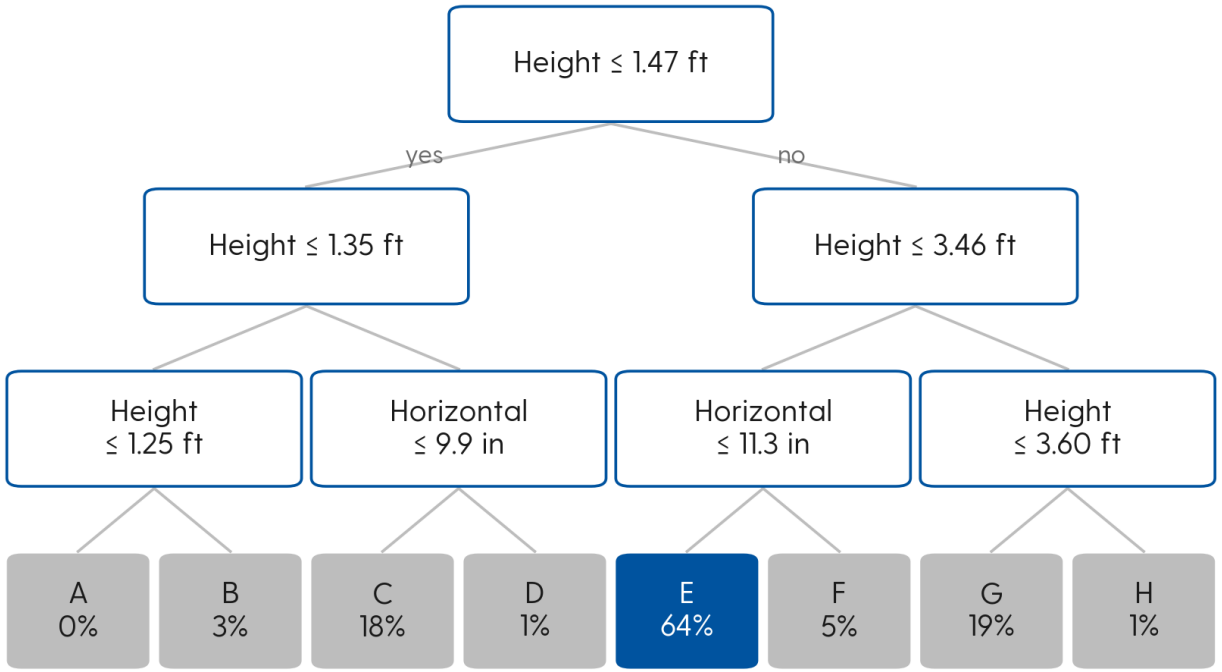


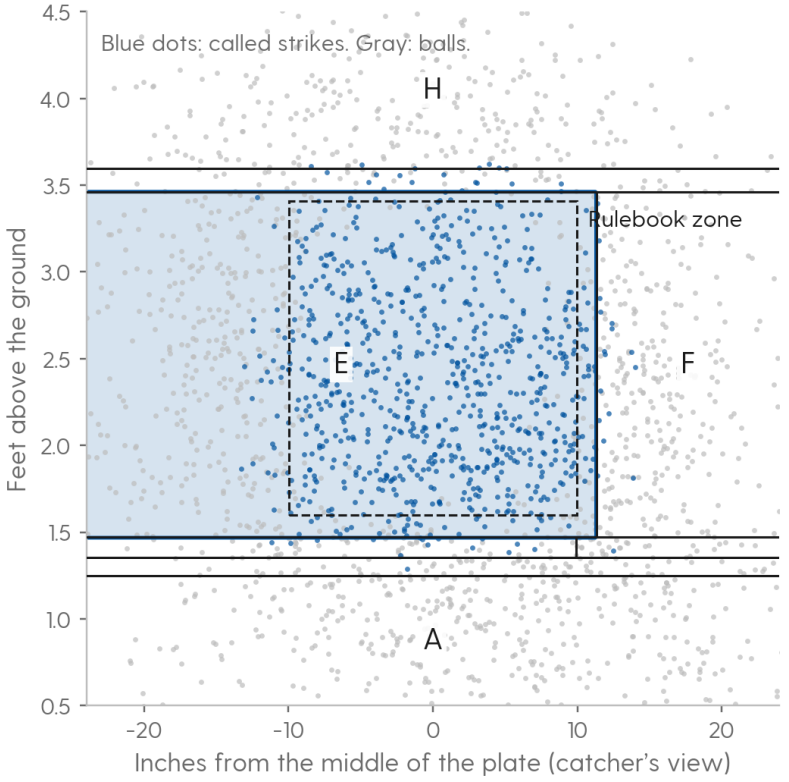


A band from the knees to the letters, with no sides: 70.8%.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

THREE LEVEL TREE STRIKE ZONE

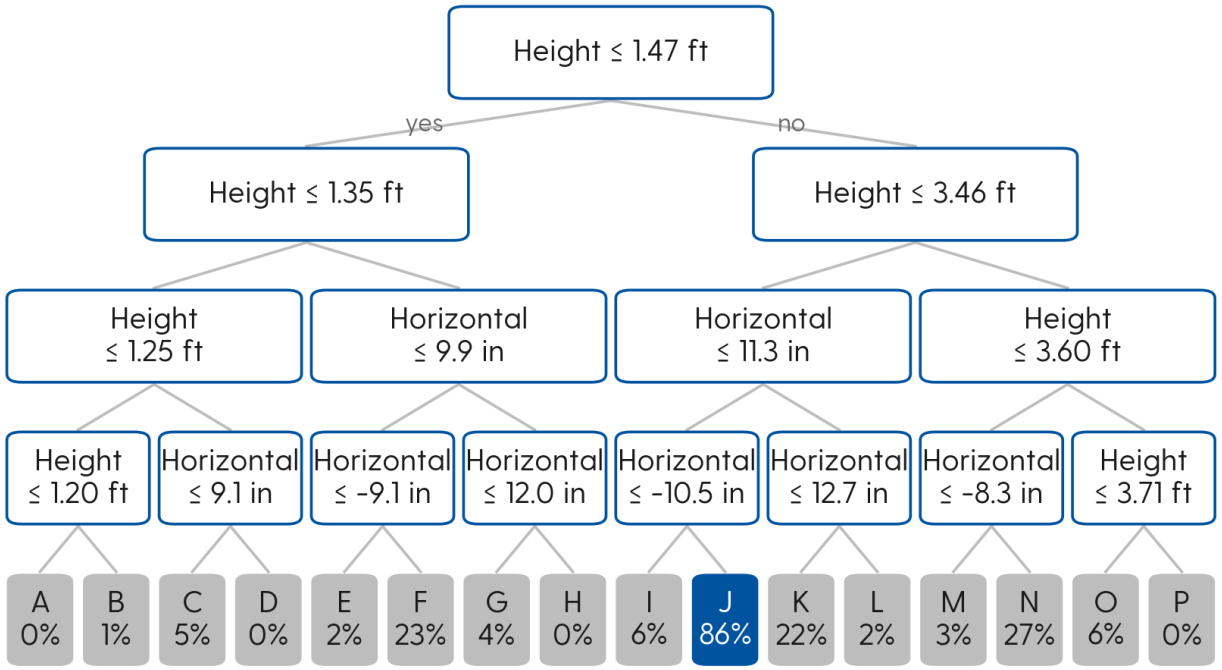


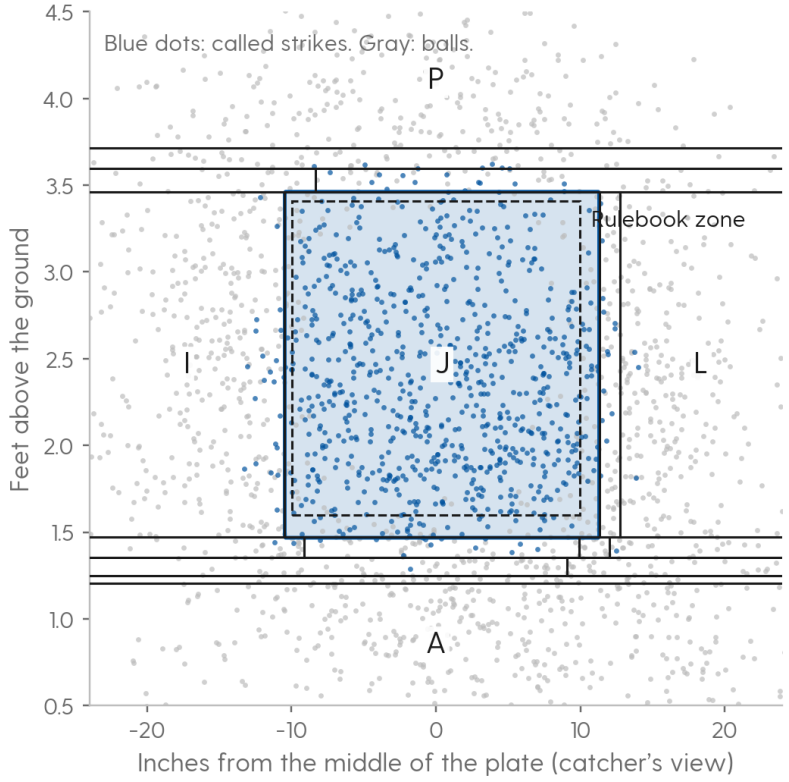


The third level finds the right edge of the plate, but not the left: 81.6%.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

FOUR LEVEL TREE STRIKE ZONE

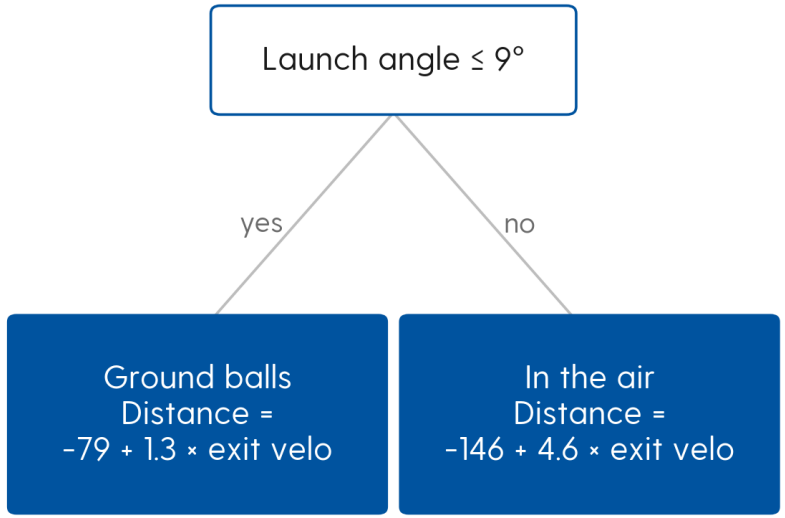


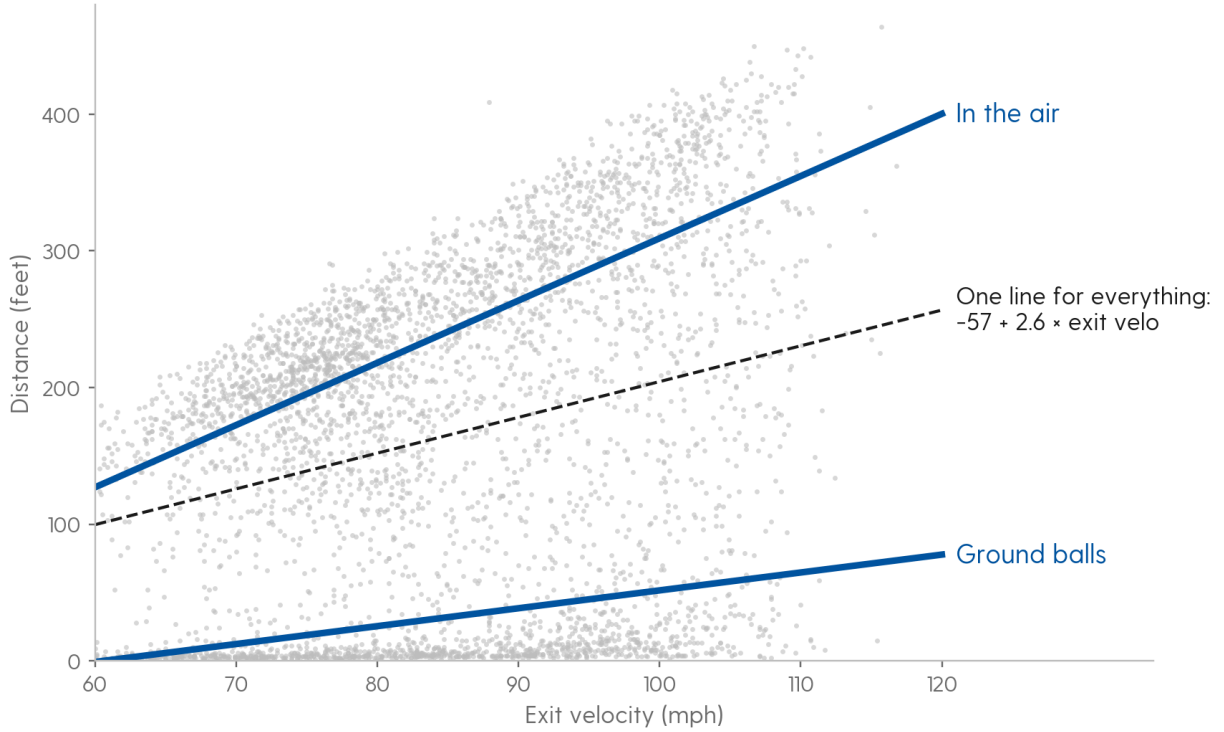


Sixteen groups draw the zone: 92.3% of new pitches called right.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

BATTED BALL DISTANCE TREE





For a single line, mph explains 8% of the variation in distance. With a tree asking a question first, mph explains 81% of the variation.

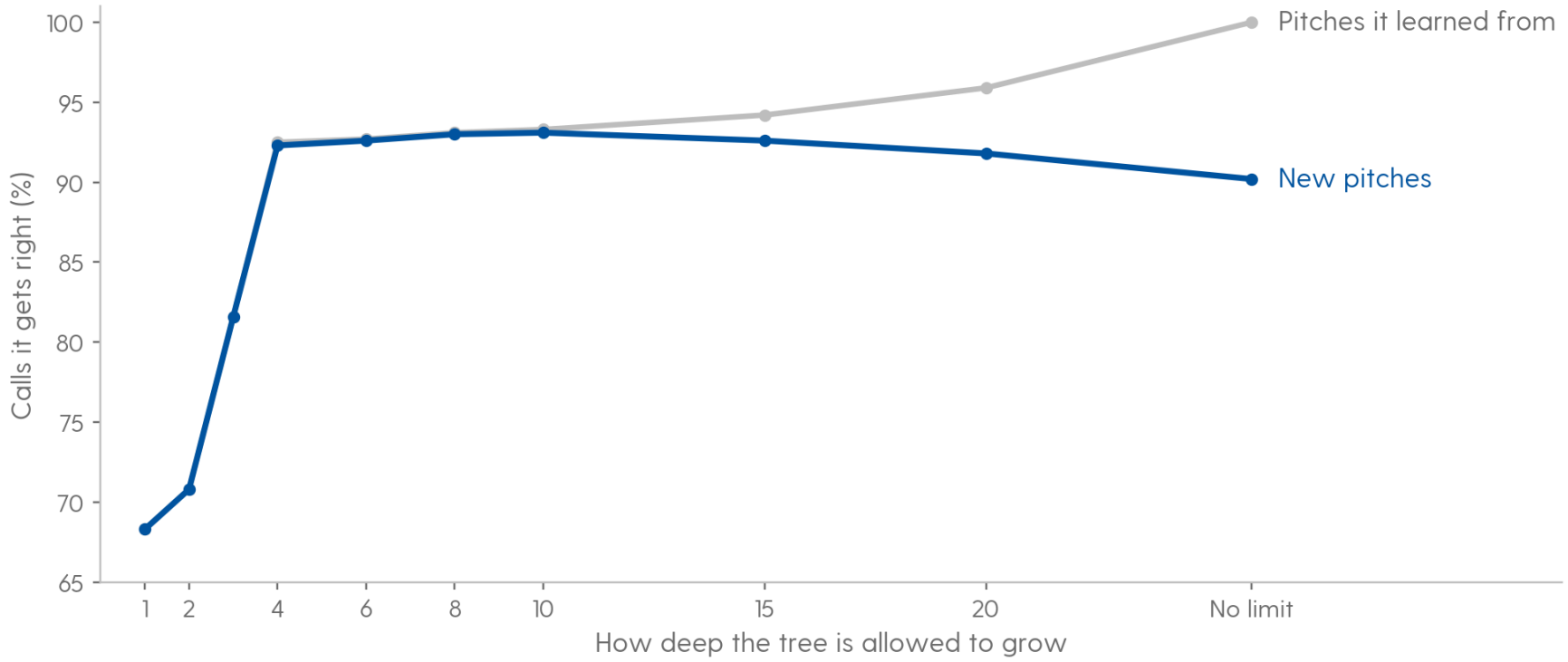
Baseball Savant (Statcast), 2024 regular season, 223,056 batted balls.

02

OVERFITTING

? WHAT WOULD OUR ACCURACY LOOK LIKE IF THE TREE KEPT SPLITTING UNTIL EVERY GROUP WAS ONE PITCH?

ACCURACY BY TREE DEPTH



The two lines match until about ten levels, then split. New pitches peak at 93.1% and fall to 90.2%.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

TRAIN/TEST SPLIT

Conducting a train/test split allows us to project how the model would perform on unseen data.

Typically, 80% of data is used for training, 20% for testing.

OVERFITTING MITIGATION

1. Increase dataset size.

2. Cross validation.

   Validate the model's accuracy by holding back portions of the training set. Use this process to determine the ideal hyperparameters, then test on unseen data.

3. Simplify the model.

   Manually tune hyperparameters (e.g.: limit the depth of a tree) or use a simpler model (e.g.: a regression instead of a neural network.). Using an ensemble model (random forest) can also help.

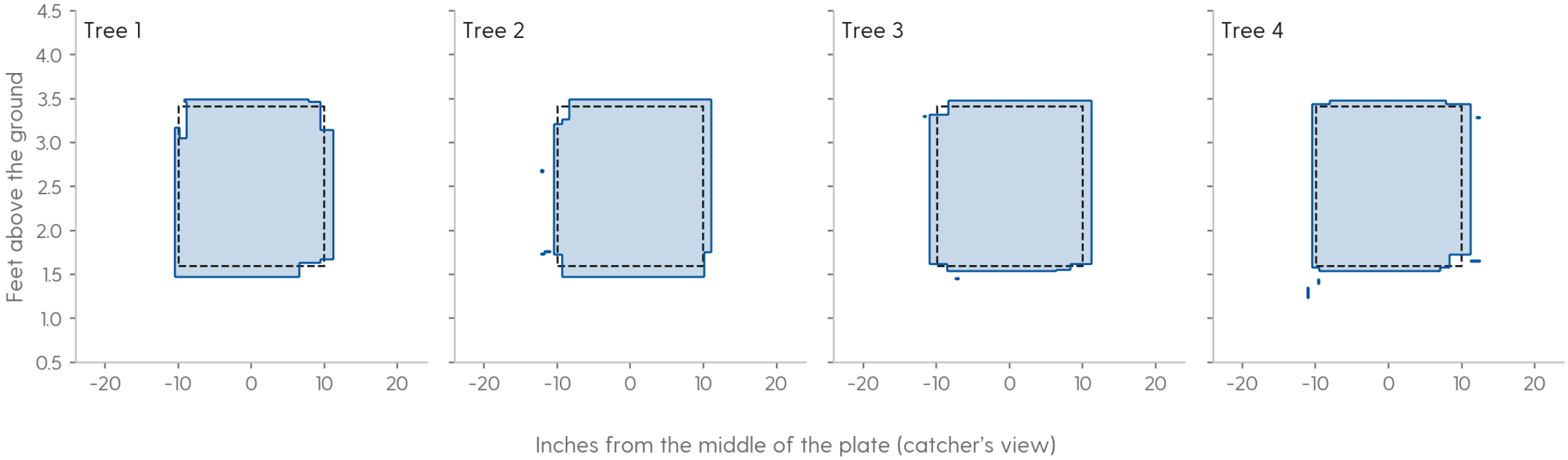
03

RANDOM FORESTS

RANDOM FOREST

1. Grow many trees, each on a random sample of the rows.
2. At each split, each tree considers a random subset of the columns.
3. The trees vote.

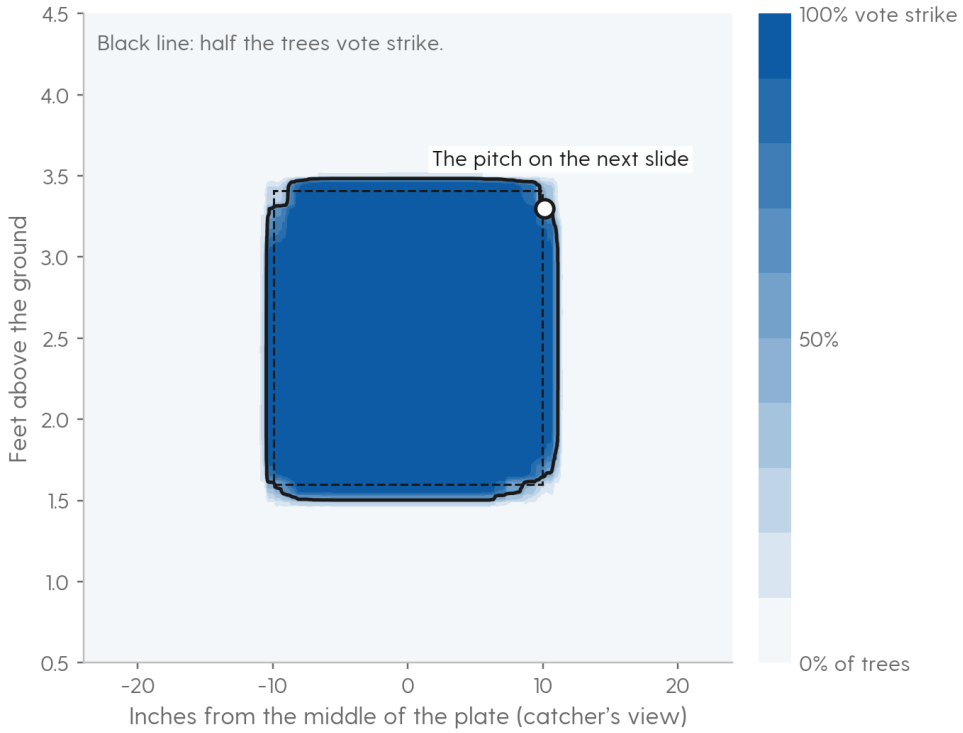
FOUR TREES FROM ONE FOREST



Each tree is eight levels deep (max_depth=8). They agree in the middle of the zone and disagree at the edges.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

THE FOREST'S VOTE



Almost every tree agrees inside and outside the zone. Only a thin band at the edges splits the vote.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

A CLOSE CALL

	THIS PITCH
Where it crossed	10.1 in right of the middle, 3.30 ft high
Batter's zone	1.49 to 3.05 ft
Trees voting strike	60 of 100
Forest's call	Strike
One eight level tree's call	Ball
Umpire's call	Ball

The forest says how sure it is: 60 of 100 means a close call. A single tree only says yes or no.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

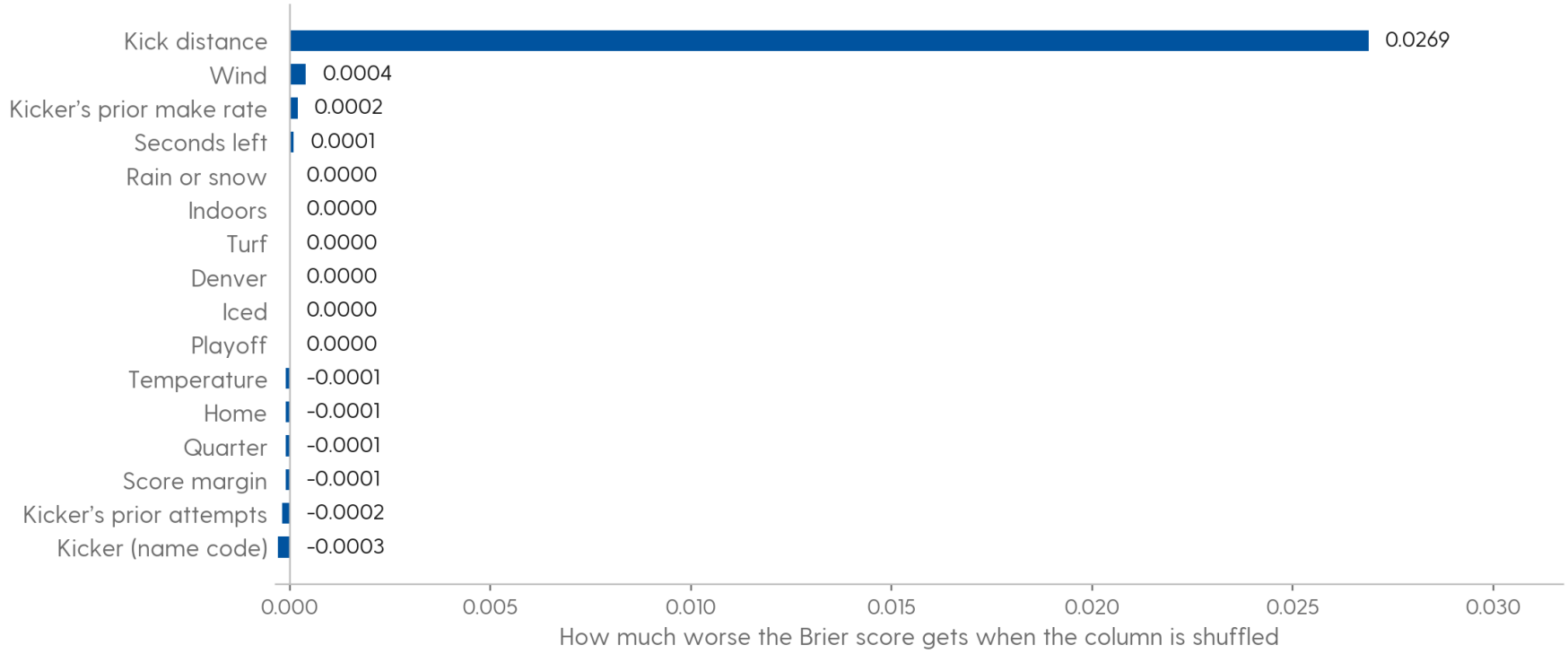
TREE VS FOREST

	SINGLE TREE	RANDOM FOREST
Can a coach read how it works?	Yes, a few yes or no questions	No, multitude of trees
Accuracy on new pitches when deep	Drops when deep (90.2%)	Holds up better (92.2%)
Probabilities	A deep tree only says 0% or 100%	Any share from 0% to 100%, even when every tree is fully grown
How you look inside	Read the questions	Permutation importance
Cost to train and run	Cheaper	More expensive

The forest predicts better. The tree is the one you can hand a coach.

Source: Baseball Savant, every taken pitch in the 2024 regular season (366,873). Trees trained on 80%, scored on the other 20%.

LOOKING INSIDE A FOREST



Shuffle kick distance and the forest's predictions get far worse. Shuffle any other column and they barely move.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

04

CONFUSION MATRIX



IS A PITCH A CALLED STRIKE? WHAT ARE THE TWO WAYS OUR MODEL CAN BE WRONG?

OPTIMIZATION GOALS

Accuracy

The model is right as often as possible. Typically the default optimization goal.

Precision

When the model predicts “positive” it’s usually correct. Minimizes false positives.

Recall

When something is actually “positive,” the model usually catches it. Minimizes false negatives.

WHAT SHOULD CANCER SCREENING BE OPTIMIZED FOR? SPAM EMAIL DETECTION? A FUMBLE VS. INCOMPLETE CALL?

Accuracy: The model is right as often as possible. Typically the default optimization goal.

Precision: When the model predicts “positive” it’s usually correct. Minimizes false positives.

Recall: When something is actually “positive,” the model usually catches it. Minimizes false negatives.

PENALIZING MISTAKES

1. A model learns by avoiding mistakes.
2. By default every mistake costs the same.

   When misses are rare, the cheapest answer is to say “good” every time.
3. You can make the mistake you care about cost more.

   The model trades some of the cheaper mistakes to avoid it.

05

CALIBRATION

CALIBRATION

1. **Probabilities should mean what they say.**

   Of the pitches a model calls 70% strikes, about 70% should be strikes.

2. **Accuracy only checks which side of 50% a prediction lands on.**

   Calibration checks the number itself.

3. **Often, the calibration matters, not just accuracy.**

   A model which is 100% accurate could tell a coach they have a 95% of making a field goal that the kicker actually has a 55% chance of making.

06

DATA LEAKAGE

DATA LEAKAGE

Using information that wouldn't be known when the prediction is made.

? IN WEEK 5, WHY DID OPPONENTS' FULL-SEASON WIN PERCENTAGE PREDICT SECOND-HALF RECORDS SO WELL?

4TH AND GOAL PROMPT

1. **Your question and audience.**

   An NFL head coach wants to know what leads to success on 4th and goal from the 1.

2. **How you define success.**

   A first down (pff_FIRST_DOWN_GAINED = 1).

3. **What data you have available and want to use.**

   PFF play data: offensive formation, defensive front, linebacker depth, run or pass, and the quarterback's drop depth.

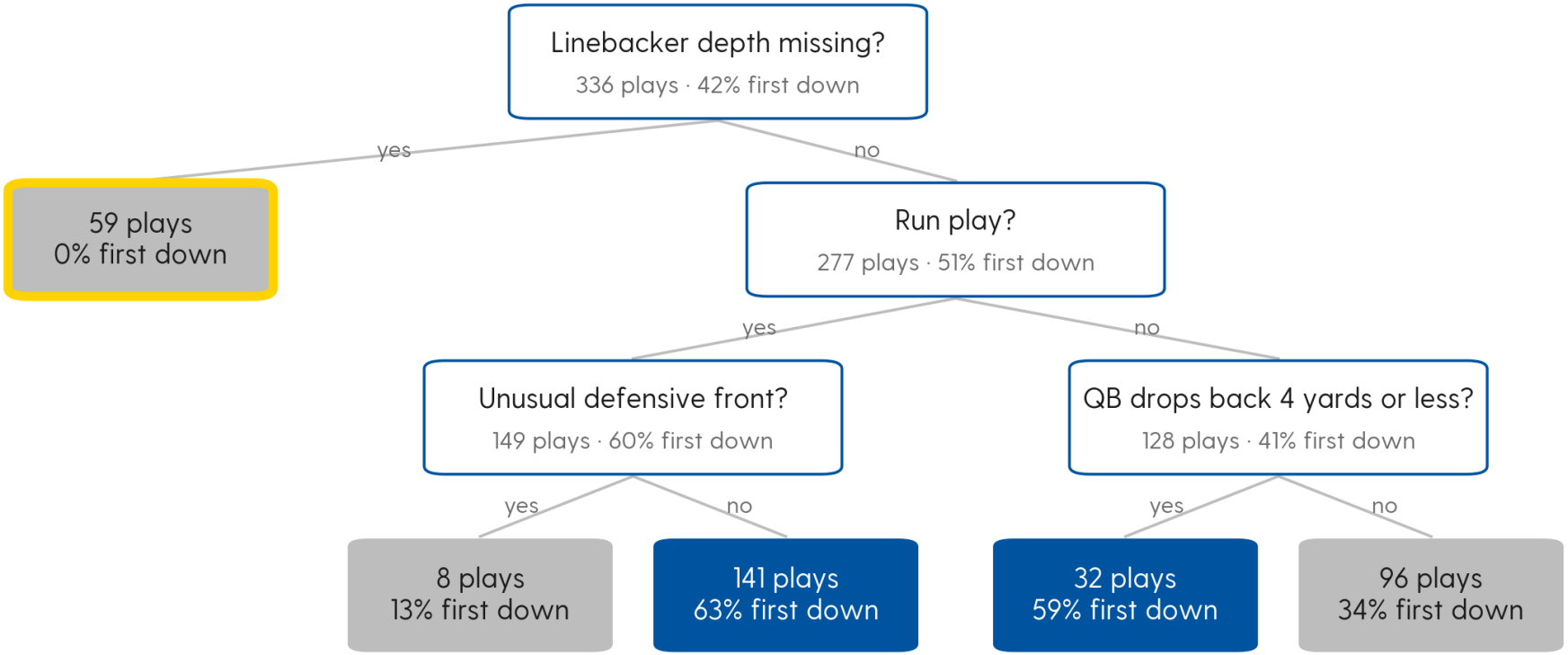
4. **What you want to optimize for.**

   Catch as many of the successful plays as possible (recall).

5. **What to watch out for.**

   Teams with better offenses may go for it more often.

4TH AND GOAL DECISION TREE



The first split is whether linebacker depth is missing. All 59 plays where it was missing failed.

PFF play data, 2025 example.



WHY WOULD EVERY PLAY MISSING LINEBACKER DEPTH FAIL?

CHECKING FOR LEAKAGE

1. Make sure that every input is known at the decision point.
2. Be suspicious of a model that is nearly perfect.
3. Look at which variables have the most importance.

07

PROMPTING THE AGENT

PROMPT ENGINEERING

1. **Your question and audience.**

   A head coach wants to know what factors lead to success on 4th and 1 to help him decide what to do when he goes for it.

2. **How you define success. Ask an LLM to grill you on edge cases.**

   A first down. (What about a touchdown? What about a defensive penalty? Offensive penalty?)

3. **What data you have available and want to use.**

   I am interested in how defensive scheme, offensive formation, and run/pass decisions all play into this. Only include NFL regular season and playoff games. I do not want personnel ratings incorporated in the analysis.

4. **What you want to optimize for.**

   I want to predict if a 4th down is successful or not as accurately as possible. (Not precision or recall).

5. **What to watch out for.**

   Teams may go for it on 4th and short more often when they're losing. We need to control for offensive and defensive strength.

VOICE MEMO CONTEXT

Okay, 4th and 1. I keep going back and forth on it. I trust the sneak, our guy's good at it. But then you see a heavy front, they stack eight, nine guys in there, and our line just, we don't move anybody. That's what kills me. Going for it at our own 35 and getting stuffed, handing them a short field. I'd rather punt than do that. And does it even matter where we are on the field? Is the 40 different from the goal line? Oh, and the ones where there's a flag and the snap doesn't count, I don't care about those. Half the time the offense is doing a hard count just trying to draw the other team offsides. I just need something I can use on the sideline.

Fictional head coach transcription.

/GOAL

/goal <condition>

Sets a finish line, and Claude starts working toward it.

A second model checks after every turn.

A small, fast model reads the conversation and decides whether the condition is met.

There is no built-in turn or time limit.

Add one: “or stop after 15 turns.”

/goal shows the status. /goal clear stops it.

VAGUE VS CONCRETE GOAL

VAGUE

/goal build an incredibly accurate 4th and 1 model

CONCRETE

/goal train a decision tree on 2016–2023 and test on 2024–2025; reach at least 72% accuracy, 3 points better than always predicting a conversion (69%); print accuracy, recall on failed attempts and the confusion matrix; list every column and confirm none is known only after the snap; or stop after 15 turns

08

THIS WEEK'S WORK

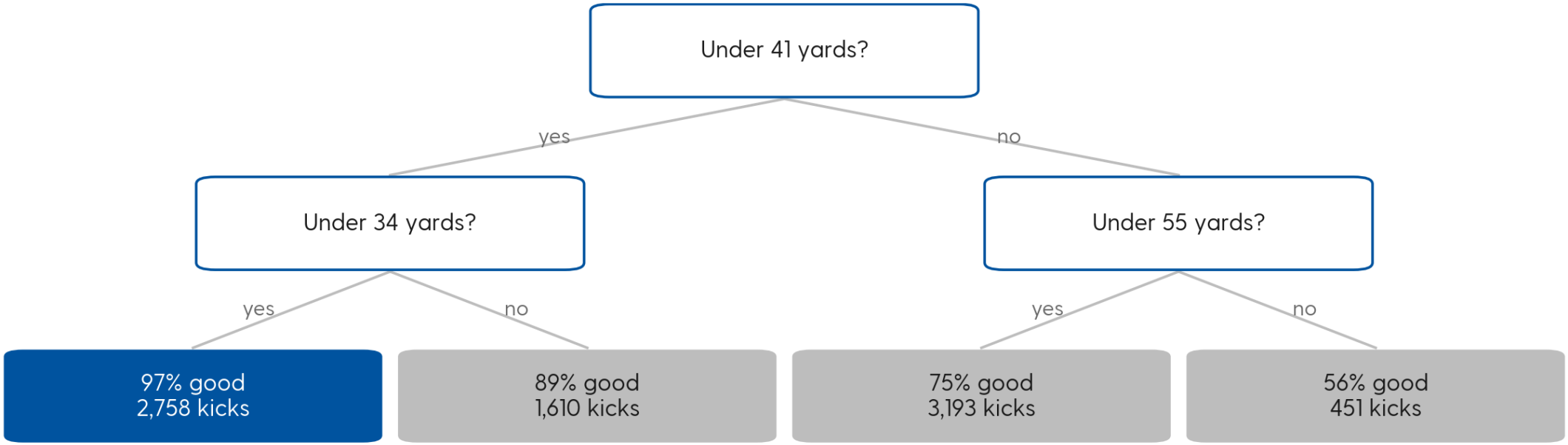
EVERY KICK IS GOOD

KICK DISTANCE	KICKS	MADE
Under 30 yards	2,311	97.7%
30 to 39 yards	2,865	92.3%
40 to 49 yards	2,966	78.4%
50 to 54 yards	1,291	70.3%
55 yards and up	582	56.0%

Saying every kick is good is right 84.8% of the time on the test kicks, but it never catches a miss.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

TWO LEVEL TREE ON KICK DISTANCE



Every box is mostly makes, so it calls every kick good: 84.8% right, and it catches none of the misses.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

THURSDAY

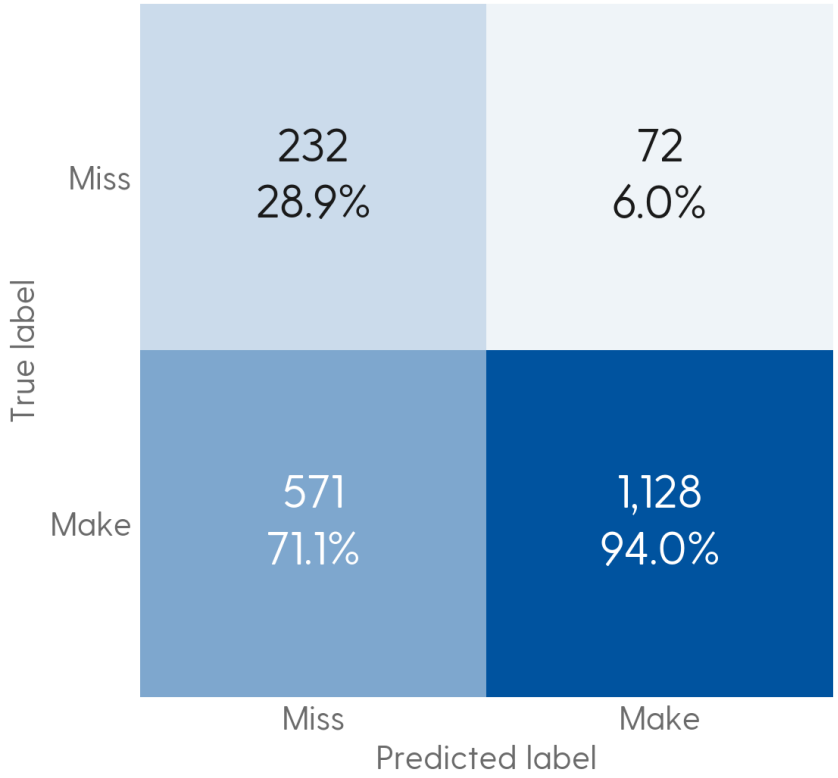
A MISSED KICK COSTS 5.5X MORE

	ACCURACY	MISSES CAUGHT	PREDICTED MISSES THAT WERE MISSES
Every mistake costs the same	84.8%	0.0%	None predicted
A missed kick costs more	67.9%	76.3%	28.9%

It predicts a miss from 38 yards out. It catches 76% of misses, and 29% of the kicks it calls misses really are.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

MISSED KICK TREE CONFUSION MATRIX



It catches 232 of the 304 misses, but 571 of the kicks it calls misses were made.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

MAKE CHANCES INSTEAD OF MAKE OR MISS

KICK DISTANCE	TRAINING KICKS	MAKE CHANCE	FLAG
Under 34 yards	2,758	97%	
34 to 40 yards	1,610	89%	
41 to 54 yards	3,193	75%	Risky
55 yards and up	451	56%	Risky

The same two level tree, read as each box's make rate instead of its yes or no. Kicks under 80% get flagged as risky, and the coach decides.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

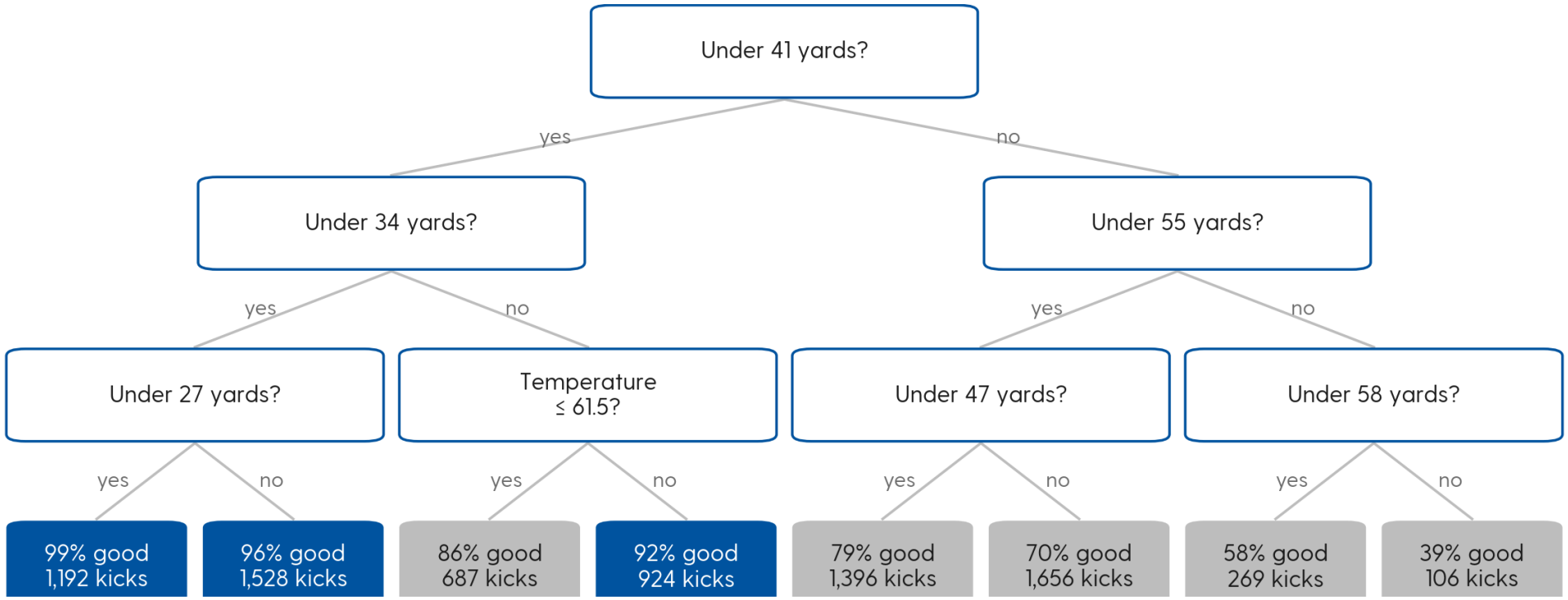
KICKER TREE WITH NO DEPTH LIMIT

	ACCURACY
Kicks it learned from	96.2%
New kicks	77.0%
Every kick is good, on new kicks	84.8%

96.2% on the kicks it learned from, 77.0% on new ones: worse than saying every kick is good.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

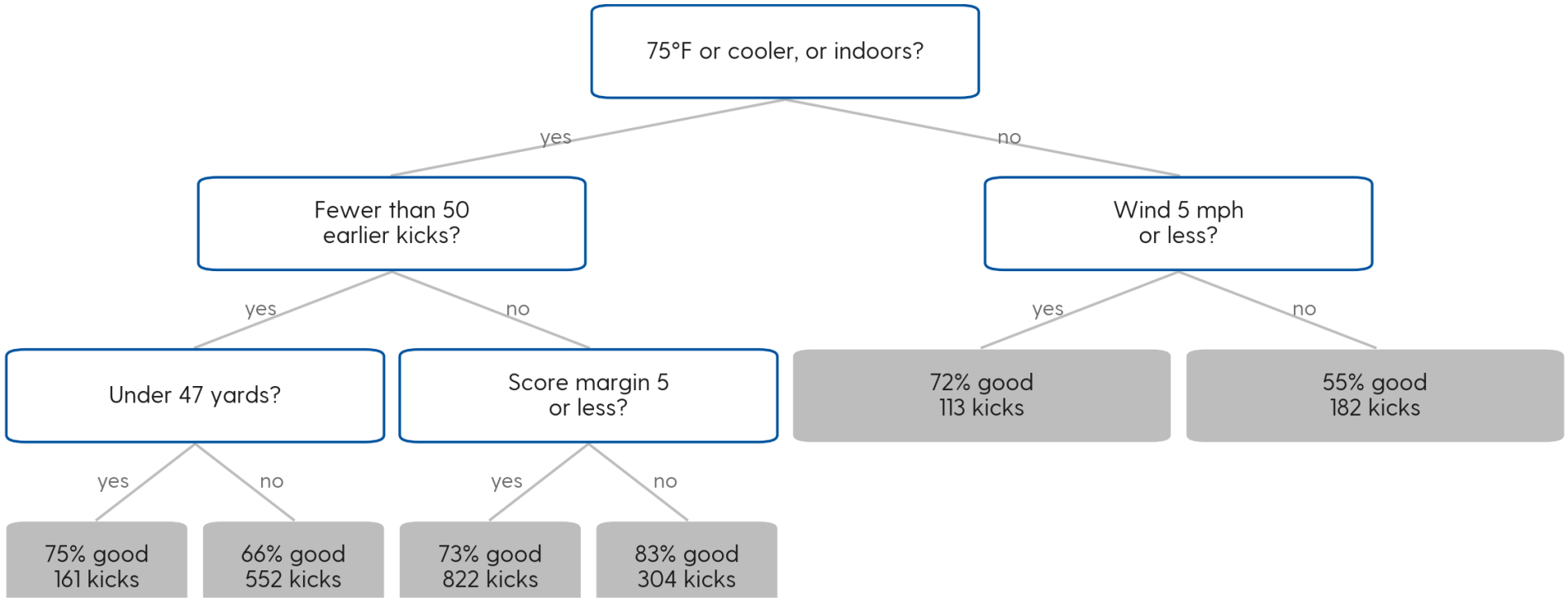
THREE LEVEL FIELD GOAL TREE



Distance takes six of the seven questions. From 34 to 40 yards, outdoors at 61°F or colder, 86.0% are good; warmer or indoors, 92.0%.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

TREE FOR 45 TO 54 YARD KICKS



Zoom in on one distance band and the splits change: temperature, the kicker's experience, the score and the wind.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).

FIELD GOAL TREE VS RANDOM FOREST

2024 AND 2025, 2,257 KICKS	ACCURACY	BRIER SCORE (LOWER IS BETTER)
Every kick good	84.9%	0.128
Three level tree	84.4%	0.118
Random forest	84.9%	0.116

Every kick good gives every kick the same make chance. The tree and the forest give long kicks lower chances, which is closer to what happened, even when the tree's yes or no is wrong more often. The forest's chances are closest.

Source: nflverse play-by-play, field goal attempts 2016 to 2025 (10,015 kicks).



IF A COACH WANTED YOUR HELP DECIDING WHEN TO KICK A FIELD GOAL, WHAT WOULD YOU BUILD FOR THEM AS A DELIVERABLE?

THURSDAY

YOUR TURN

Build the best field goal make-probability model you can.

Train on 2016 to 2023, test on 2024 and 2025.

Try setting a /goal.

A concrete success condition and a turn cap. Save it as outputs/lab/goal.md and commit it before you run it.

Report on the test kicks.

What metrics (accuracy, calibration, etc) will you use to determine whether it is useful?

Create a chart you could hand to a head coach to assist with their decision making process.

Data: weeks/week12/data/nfl-field-goals-2016-2025.csv after /update: 10,015 field goal attempts, 2016 to 2025.

BEFORE YOU PUSH ON MONDAY, NOVEMBER 16

Commit prediction.md, goal.md, your field goal model and calibration chart, your code, and your coach page.

Answer every question in BRIEF.md, in your own words.

Run /audit, then /quiz-me.

Upload your repository's link to Canvas by Monday, November 16 at 11:59pm.